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A SOLVABLE CASE OF THE QUINTIC EQUATION.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XVIII. (1882), pp. 154–157.]

The roots of the general quintic equation

$$(a, b, c, d, e, f \mid x, 1)^5 = 0$$

may be taken to be

$$\begin{aligned} & -\frac{b}{a} + B + C + D + E, \\ & -\frac{b}{a} + \omega B + \omega^2 C + \omega^3 D + \omega^4 E, \\ & -\frac{b}{a} + \omega^2 B + \omega^4 C + \omega D + \omega^3 E, \\ & -\frac{b}{a} + \omega^3 B + \omega C + \omega^4 D + \omega^2 E, \\ & -\frac{b}{a} + \omega^4 B + \omega^3 C + \omega^2 D + \omega E, \end{aligned}$$

where ω is an imaginary fifth root of unity; and if one of the four functions B, C, D, E is = 0, say if $E = 0$ (this implies of course a single relation between the coefficients), then the equation is solvable.

Writing $x = \xi - \frac{b}{a}$, we have

$$(a, b, c, d, e, f \mid \xi - \frac{b}{a}, 1)^5 = (a', 0, c', d', e', f' \mid \xi, 1)^5,$$

where

$$\begin{aligned} a' &= a, \\ ac' &= ac - b^2, \\ a^2 d' &= a^2 d - 3abc + 2b^3, \\ a^3 e' &= a^3 e - 4a^2 bd + 6ab^2 c - 3b^4, \\ a^4 f' &= a^4 f - 5a^3 be + 10a^2 b^2 d - 10ab^3 c + 4b^5, \end{aligned}$$

and the roots of the new equation

$$(a', 0, c', d', e', f' \mid \xi, 1)^5 = 0$$

have the above-mentioned values, omitting therefrom the term $-\frac{b}{a}$: we find without difficulty

$$\begin{aligned} 2\frac{c'}{a} &= -BE - CD, \\ 2\frac{d'}{a} &= -B^2 D - BC^2 - CE^2 - D^2 E, \\ \frac{e'}{a} &= -B^3 C - B^2 E^2 + BCDE + BD^3 + C^3 E + C^2 D^2 - DE^3, \\ \frac{f'}{a} &= -B^5 + 5B^2 DE - 5B^2 C^2 E - 5B^2 CD^2 + 5BC^3 D + 5BCE^3 \\ &\quad - 5BD^3 E^2 - C^5 + 5CD^3 E - 5CD^2 E^2 - D^5 - E^5, \end{aligned}$$

and hence, when $E = 0$, we have

$$\begin{aligned} 2\frac{c'}{a} &= -CD, \\ 2\frac{d'}{a} &= -B^2 D - BC^2, \end{aligned}$$

$$\begin{aligned}\frac{e'}{a} &= -B^3C - BD^3 - C^2D^2, \\ \frac{f'}{a} &= -B^5 - 5B^2CD^2 + 5BC^3D - C^5 - D^5,\end{aligned}$$

or, as these may be written,

$$\begin{aligned}-2\frac{c'}{a} &= CD, \\ -2\frac{d'}{a} &= B^2D + BC^2, \\ -\frac{e'}{a} - \frac{c'^2}{a^2} &= B^3C - BD^3, \\ -\frac{f'}{a} &= B^5 + C^5 + D^5 - 10\frac{c'}{a}(B^2D - BC^2),\end{aligned}$$

equations which imply a single relation between the coefficients a' , c' , d' , e' , f' . Supposing this satisfied, we may attend only to the first three equations; or, writing for convenience,

$$\begin{aligned}\gamma &= -2\frac{c'}{a} = -\frac{2}{a^2}(ac - b^2), \\ \delta &= -2\frac{d'}{a} = -\frac{2}{a^3}(a^2d - 3abc + 2b^3), \\ \theta &= -\frac{e'}{a} - \frac{c'^2}{a^2} = -\frac{1}{a^4}\{a^2(ae - 4bd + 3c^2) + (ac - b^2)^2\},\end{aligned}$$

the equations are

$$\begin{aligned}\gamma &= CD, \\ \delta &= B(BD + C^2), \\ \theta &= B(D^3 - B^2C).\end{aligned}$$

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The first equation gives $C = \frac{\gamma}{D}$, and substituting this value in the other two equations, we have

$$\begin{aligned}B^2D^2 + B\gamma^2 - \delta D &= 0, \\ B^3\gamma^2 + BD^4 + \theta D &= 0.\end{aligned}$$

Eliminating B , the result is obtained in the form $\det. = 0$, where in the last column of the determinant each term is divisible by D ; and omitting this factor, the result is

$$\begin{vmatrix} D^2, & \gamma^2, & 0, & -\delta D \\ 0, & D^2, & \gamma^2, & -\delta D \\ \gamma^2, & 0, & D^4, & \theta \\ 0, & \gamma^2, & 0, & D^4, \quad \theta \end{vmatrix} = 0.$$

If, in order to develop the determinant, we consider it as a sum of products, each first factor being a minor composed out of columns 1 and 2, and the second factor being the complementary minor composed out of columns 3, 4, 5 (the several products being of course taken each with its proper sign), the expansion presents itself in the form

$$\begin{aligned}& D^4(-\gamma^2\theta D^2 + \theta^2 D^3) \\ & - D^2\gamma^2(-\theta\gamma^2 D^2 + \theta D^5 - \delta^2 D^3) \\ & - \gamma^2 D^2(-\delta D^6 - \theta\gamma^4) \\ & + \gamma^4(-\gamma^4\delta D^2 - \delta^3 D^4 - \delta\gamma^2 D^4) \\ & - \gamma^8 D^2.\end{aligned}$$

Hence, collecting, and changing the sign of the whole expression, we obtain

$$BD^6 - (2\gamma^2\delta + \theta^2)D^4 + (-\gamma^4\delta + \delta^2\gamma^2\theta + \delta^2)D^3 + \gamma^8 = 0,$$

viz. a cubic equation for D^2 . We have then as above $C = \frac{\gamma}{D}$, and B is given rationally as the common root of the foregoing quadric and cubic equations satisfied by B .

Substituting for γ, δ, θ their values in terms of the original coefficients, the equation for D^2 becomes

$$\begin{cases} 2(a^2d - 3abc + 2b^3)(aD)^6 \\ + a^2[(ae - 4bd + 3c^2)^2a^2 + a^2(ac - b^2)^2(ae - 4bd + 3c^2) - 16(ac - b^2)^3(a^2d - 3abc + 2b^3)](aD)^4 \\ + [28(ac - b^2)^2(a^2d - 3abc + 2b^3) \\ + 4(ae - b^2)^2\{4a^2(ac - b^2)(a^2d - 3abc + 2b^3)(ae - 4bd + 3c^2) + 8(a^2d - 3abc + 2b^3)^2\}](aD)^3 \\ - 128(ac - b^2)^3\{a^2(ae - 4bd + 3c^2)(ac^2 - b^2)^2\} = 0, \end{cases}$$

and the solution of the given quintic equation thus ultimately depends upon that of this cubic equation.

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[ADDITION TO MR HUDSON'S PAPER "ON EQUAL ROOTS OF EQUATIONS."]

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XVIII. (1882), pp. 226–229.]

It seems desirable to present in a more developed form some of the results of the foregoing paper.

Thus, if the equation $(a_0, a_1, \dots, a_n \mid x, 1)^n = 0$ of the order n has $n - v$ equal roots, where v is not $> \frac{1}{2}n - 1$, then we have $\psi(r, v + 1, m) = 0$, where m has any one of the values $0, 1, \dots, n - 2v - 2$, and r any one of the values

$$2v + 2, 2v + 3, \dots, n - m.$$

The signification is

$$\begin{aligned} \psi(r, v + 1, m) = & r \cdot 1 \cdot \frac{1}{[r]^{v+1}} a_m a_{r+m} \\ & - (r - 2) \cdot \frac{v + 1}{1} \cdot \frac{1}{[r - 1]^{v+1}} a_{m+1} a_{r+m-1} \\ & + (r - 4) \cdot \frac{v + 1 \cdot v + 2}{1 \cdot 2} \cdot \frac{1}{[r - 2]^{v+1}} a_{m+2} a_{r+m-2} \\ & + (-)^s (r - 2s) \cdot \frac{[v + 1]^s}{[s]^s} \cdot \frac{1}{[r - s]^{v+1}} a_{m+s} a_{r+m-s} \\ & \dots \\ & + (-)^{v+1} (r - 2v - 2) \cdot \frac{1}{[r - v - 1]^{v+1}} a_{m+v+1} a_{r+m-v-1}. \end{aligned}$$

Thus, when $v = 0$, the condition is

$$r \cdot \frac{1}{r \cdot r - 1} a_m a_{r+m} - (r - 2) \cdot \frac{1}{r - 1 \cdot r - 2} a_{m+1} a_{r+m-1} = 0,$$

that is,

$$a_m a_{r+m} - a_{m+1} a_{r+m-1} = 0,$$

satisfied when the equation has all its roots equal.

The values of m are $0, 1, 2, \dots, n - 2$, and those of r are $2, 3, \dots, n - m$; in particular, if $m = 0$, the values of r are $2, 3, \dots, n$, and the corresponding conditions are

$$\begin{aligned} a_0 a_2 - a_1^2 &= 0, \\ a_0 a_3 - a_1 a_2 &= 0, \\ a_0 a_n - a_1 a_{n-1} &= 0, \end{aligned}$$

and so for the different values of m up to the final value $r = 2$, for which $m = n - 2$, and the condition is

$$a_{n-2} a_n - a_{n-1}^2 = 0;$$

we have thus, it is clear, the whole series of conditions included in

$$\begin{vmatrix} a_0, & a_1, & a_2, & \dots, & a_{n-1} \\ a_1, & a_2, & a_3, & \dots, & a_n \end{vmatrix} = 0,$$

which are obviously satisfied in the case in question of the roots being all equal.

Again, when $v = 1$, the condition for $n - 1$ equal roots is

$$r \cdot 1 \cdot \frac{1}{r \cdot r - 1 \cdot r - 2} a_m a_{r+m} - (r - 2) \cdot \frac{2}{1} \cdot \frac{1}{r - 1 \cdot r - 2 \cdot r - 3} a_{m+1} a_{r+m-1} + (r - 4) \cdot 1 \cdot \frac{1}{r - 2 \cdot r - 3 \cdot r - 4} a_{m+2} a_{r+m-2} = 0;$$

that is,

$$r - 1 \cdot r - 2 - r - 1 \cdot r - 3 + r - 2 \cdot r - 3 = 0;$$

or, what is the same thing,

$$(r - 3) a_m a_{r+m} - 2(r - 2) a_{m+1} a_{r+m-1} + (r - 1) a_{m+2} a_{r+m-2} = 0,$$

where $n = 4$ at least, and m, r have the values

$$m = 0, 1, 2, \dots, n - 4$$

$$r = \begin{cases} 4, 4, \dots, 4 \\ 5, 5 \\ \vdots \\ n - 1 \\ n \end{cases}$$

thus, when $n = 4$, the only values are $m = 0, r = 4$, and the condition is

$$a_0 a_4 - 4a_1 a_3 + 3a_2^2 = 0.$$

Similarly, when $v = 2$, the condition for $n - 2$ equal roots is found to be

$$\frac{a_m a_{r+m}}{r - 1 \cdot r - 2 \cdot r - 3} - \frac{3a_{m+1} a_{r+m-1}}{r - 1 \cdot r - 2 \cdot r - 4} + \frac{3a_{m+2} a_{r+m-2}}{r - 2 \cdot r - 3 \cdot r - 5} - \frac{a_{m+3} a_{r+m-3}}{r - 3 \cdot r - 4 \cdot r - 5} = 0;$$

or, what is the same thing,

$$\begin{aligned} & r - 4 \cdot r - 5 \cdot a_m a_{r+m} \\ & - 3 \cdot r - 2 \cdot r - 5 \cdot a_{m+1} a_{r+m-1} \\ & + 3 \cdot r - 1 \cdot r - 4 \cdot a_{m+2} a_{r+m-2} \\ & - r - 1 \cdot r - 2 \cdot a_{m+3} a_{r+m-3} = 0, \end{aligned}$$

where $n = 6$ at least, and m, r have the values

$$m = 0, 1, \dots, n - 6$$

$$r = \begin{cases} 6, 6, \dots, 6 \\ 7, 7 \\ \vdots \\ n - 1 \\ n \end{cases}$$

Observe that the sum of the coefficients is 0, *viz.*

$$(r - 4)(r - 5) - 3(r - 2)(r - 5) + 3(r - 1)(r - 4) - (r - 1)(r - 2) = 0,$$

this should obviously be the case, since the conditions for $n - 2$ equal roots must be satisfied when the roots are all of them equal; and the property serves as a verification.

It is to be remarked that the equation $\psi(r, v + 1, m) = 0$ does not in all cases give all the conditions for the existence of $n - v$ equal roots in an equation of the order n ; thus when $n = 3$ and $v = 1$, we cannot by means of it obtain the condition that a cubic equation may have 2 equal roots. The problem really considered is that of the determination of those quadric functions of the coefficients which vanish in the case of $n - v$ equal roots; and in the case in question ($n = 3, v = 1$) there is no quadric function which vanishes, but the condition depends on a cubic function.

The question of the quadric functions which vanish in the case of $n - v$ equal roots, and to a small extent that of the cubic functions which thus vanish, is considered in Dr Salmon's "Note on the conditions that an equation may have equal roots," *Camb. and Dublin Math. Jour.*, t. v. (1850), pp. 159–165, and in particular the equation there obtained p. 161 is the equation $\psi(0, v + 1, n) = 0$.

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[NOTE ON MR JEFFERY'S PAPER "ON CERTAIN QUARTIC CURVES WHICH HAVE A CUSP AT INFINITY, WHEREAT THE LINE AT INFINITY IS A TANGENT."]

[From the *Proceedings of the London Mathematical Society*, vol. XIV. (1883), p. 85.]

The assumed form $\kappa\phi^2 = u_4$, or, as this is afterwards written,

$$2y^3y = ax^2 + 2bxy + cy^2 + 2ex + 2dy + \lambda,$$

is, I think, introduced without a proper explanation. Say, the form is $x^2y = z^2(* | x, y, z)^2$, it ought to be shown how for a cuspidal quartic we arrive at this form; *viz.* taking the cusp to be at the point $(x = 0, z = 0)$, $z = 0$ for the tangent at the cusp, and $x = 0$ an arbitrary line through the cusp; then the line $z = 0$ besides intersects the curve in a single point, and, if $y = 0$ is taken as the tangent at that point, the equation of the curve must, it can be seen, be of the form

$$(ax^2 + exz)y = z^2(a, b, c, f, g, h | x, y, z)^2.$$

The conic $(a, b, c, f, g, h | x, y, z)^2$ touches the quartic at each of the two intersections of the quartic with the arbitrary line $x = 0$; and we cannot, so long as the line remains arbitrary, find a conic which shall osculate the quartic at the two points in question; but, for the particular line $x + \frac{1}{2}dz = 0$, there exists such a conic, *viz.* writing x instead of $x + \frac{1}{2}dz$, the form is $x^2y = z^2(a', b', c', f', g', h' | x, y, z)^2$, and the new conic $(a', \dots | x, y, z)^2 = 0$ has the property in question. This is the adopted form, and it thus appears that in it the line $x = 0$ is a determinate line, *viz.* the line passing through the cusp and the two points of osculation of the osculating conic.

It thus appears that in the assumed form the lines $x = 0, y = 0, z = 0$ are determinate lines.

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[ADDITION TO MR HAMMOND'S PAPER "NOTE ON AN EXCEPTIONAL CASE IN WHICH THE FUNDAMENTAL POSTULATE OF PROFESSOR SYLVESTER'S THEORY OF TAMISAGE FAILS."]

[From the *Proceedings of the London Mathematical Society*, vol. XIV. (1883), pp. 88–91. Read Dec. 14, 1882.]

The extreme importance of Mr Hammond's result, as regards the entire subject of Covariants, leads me to reproduce his investigation in the notation of my Memoirs on Quantics, and with a somewhat different arrangement of the formulæ. For the binary seventhic

$$(a, b, c, d, e, f, g, h | x, y)^7,$$

the four composite seminvariants of the deg-weight 5.11 (sources of covariants of the deg-order 5.13) are

I.	II.	III.	IV.
1.7	4.6	2.10	3.3 Deg-order.
1.0	4.11	2.2	3.9 Deg-weight.
$a + 1$	$a^2eh + 1$ $a^2fg - 1$ $abdh - 4$ $abeg - 2$ $abf^2 + 6$ $ac^2h - 3$ $acd g - 2$ $acef - 6$ $ad^2f + 10$ $ad^2e - 5$ $ade^2 + \dots$ $b^2dg + 20$ $b^2ef + 57$ $bc^2g - 15$ $bcd f - 24$ $bce^2 - 30$ $bc^2e - 10$ $c^4f + 27$ $c^3de - 45$ $cd^3 + 20$	$ac + 1$ $b^2 - 1$	$ach + 2$ $adg - 7$ $ae f + 5$ $ac^2fh - 2$ $beg + 7$ $bdf + 22$ $be^2 - 25$ $cf - 27$ $cde + 45$ $d^3 - 20$

II. (continued)	III. (continued)	IV. (continued)
2.6	3.7	3.11 Deg-order.
2.4	3.1	3.5 Deg-weight.
$ae + 1$ $bd - 4$ $c^2 + 3$	$a^2h + 1$ $abg - 7$ $acf + 9$ $ade - 5$ $b^2f + 12$ $bce - 30$ $bd^2 + 20$	$ag + 1$ $bf - 6$ $ce + 16$ $d^2 - 10$
$a^2f + 1$ $abe - 5$ $ad - 2$ $ab^2d - 3$ $be^2 - 6$		

and it is here at once obvious that there exists a syzygy of the form $I = III - IV$; in fact, if in III. and IV. we write $a = 0$, then the values are each

$$= -2b(2bd - 3c^2)(6bf - 15ce + 10d^2)$$

hence $III - IV$ must divide by a , the quotient being a seminvariant of the deg-weight 4.11, which can only be a numerical multiple of the second factor of I., and is in fact this second factor, that is, we have the syzygy $I = III - IV$.

Working out the values of the four products, and joining to them the expression for the irreducible seminvariant of the same deg-weight 5.11 (Ω , Θ^* of my tables [774] for the binary sextic), we have the table:

5 . 10	5 . 11	I.	III.	IV.	II.
a^2dh	a^2efh	+1	+1		
a^2eg	a^2fg	−1	−4	+1	
a^2f^2	a^2bdh	−4	−7		
	a^2beg	−2	+3	−5	
a^2bch	a^2bf^2	+3	+9	−6	
a^2bdg	a^2c^2h	−2	−5	+2	
a^2bef	a^2cdg	−1	+28	+15	+2
a^2c^2g	a^2cef	+10	+12	−10	−7
a^2cdf	a^2d^2f	+3			+5
a^2ce^2	a^2de^2	−5	−21		−4
a^2d^2e	ab^2ch	−2	−36		
ab^3h	ab^2cg	+20	−30	+8	+7
ab^2cg	ab^2df	+1	+67	−45	−6
ab^2df	ab^2e^2	−15	+40		
ab^2e^2	abc^2g	−24	+27	−6	+7
abc^2f	abc^2f	−30	−15	−12	+22
$abcde$	$abcdf$	+27	−48		
abd^3	$abce^2$	−14	−45		−25
ac^3e	abd^2e	+20	+36	+50	−27
ac^2d^2	ac^3f	+11	+120	+30	+46
a^2c^2g	ac^2de	+1	−80	−20	−20
ab^2cf	acd^3	+9	−90		−2
ab^2de	a^2b^2h	−14	+60	−48	−7
ab^2c^2e	b^3cg	+6		−22	+26
ab^2cd^2	b^3df	+8		+86	+27
abc^3d	b^3e^2	−9		+120	−45
	b^2c^2f	−6		−80	+20
	b^2cde	+16		−90	
	b^2d^3	−8		+60	
	bc^3e	−3			
	bc^2d^2	+2			
	c^4d				

I have prefixed to the table the literal terms of the deg-weight 5 . 10; for the deg-weights 5 . 11 and 5 . 10, the numbers of terms are = 30 and 26 respectively; and it is the difference of these $30 - 26 = 4$, which gives the number of aszygetic seminvariants of the deg-weight 5 . 11.

781.

ON THE AUTOMORPHIC TRANSFORMATION OF THE BINARY CUBIC FUNCTION.

[From the *Proceedings of the London Mathematical Society*, vol. XIV. (1883), pp. 103–108. Read Jan. 11, 1883.]

I CONSIDER the cubic equation $(a, b, c, d \mid x, 1)^3 = 0$. It is shown (Serret, *Cours d'Algèbre supérieure*, 4th ed., Paris, 1879, t. II. pp. 466–471) how, given one root of the equation, the other two roots can be each of them expressed rationally in terms of this root and of the square root of the discriminant; viz. making the proper changes of notation, and writing

$$\begin{aligned}
 A, B, C &= ac - b^2, \quad ad - bc, \quad bd - c^2, \quad \lambda = \sqrt{-1}, \\
 \Omega &= B^2 - 4AC, \quad = a^2d^2 - 4ac^3 + 6abcd - 4b^3d - 3b^2c^2, \\
 \alpha &= \frac{-\lambda\sqrt{\Omega} + B}{2\lambda\sqrt{\Omega}}, \quad \beta = \frac{+2C}{2\lambda\sqrt{\Omega}}, \quad \gamma = \frac{-2A}{2\lambda\sqrt{\Omega}}, \quad \delta = \frac{-\lambda\sqrt{\Omega} - B}{2\lambda\sqrt{\Omega}},
 \end{aligned}$$

(values which give $\alpha + \delta = -1$, $\alpha\delta - \beta\gamma = -1$, and therefore also

$$\alpha^2 + \alpha\delta + \delta^2 + \beta\gamma = 0,$$

which is the condition in order that the function $\phi x = \frac{\alpha x + \beta}{\gamma x + \delta}$ may be periodic of the third order, $\phi^3 x = x$), then, u being a root of the equation, say $(a, b, c, d \mid u, 1)^3 = 0$, the other two roots are

$$\phi u = \frac{\alpha u + \beta}{\gamma u + \delta},$$

and

$$\phi^2 u = \phi^{-1} u = \frac{(\alpha^2 + \beta\gamma)u + \beta(\alpha + \delta)}{\gamma(\alpha + \delta)u + \delta^2 + \beta\gamma} = \frac{-\delta u - \beta}{-\gamma u + \alpha},$$

where observe that, by the change of $\sqrt{\Omega}$ into $-\sqrt{\Omega}$, $\alpha, \beta, \gamma, \delta$ become $\delta, -\beta, -\gamma, \alpha$; so that the last-mentioned value $\phi^{-1}u$ is, in fact, the value obtained from ϕu by the mere change of sign of the radical.

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It is to be observed that, if we have between two roots u, v of the equation

$$(a, b, c, d \mid x, 1)^3 = 0,$$

a relation $v = \frac{\alpha u + \beta}{\gamma u + \delta}$, where $\alpha, \beta, \gamma, \delta$ have given values, this implies in the first place a relation between a, b, c, d (and the given values of $\alpha, \beta, \gamma, \delta$), and it implies more-over that u , and consequently also v , are not any roots whatever, but two determinate roots of the equation; viz. u, v will be each of them expressible rationally in terms of a, b, c, d and $\alpha, \beta, \gamma, \delta$. And if, in order that (a, b, c, d) may remain arbitrary, we consider $\alpha, \beta, \gamma, \delta$ as given quantities satisfying the relation which exists between these quantities and (a, b, c, d) , then in general we still have u, v determinate roots of the cubic equation. But in the foregoing solution u is any root whatever of the cubic equation.

To examine how this is, starting from the equations

$$(a, b, c, d \mid u, 1)^3 = 0, \quad (a, b, c, d \mid v, 1)^3 = 0, \quad v = \frac{\alpha u + \beta}{\gamma u + \delta},$$

we have

$$a(u^3 - v^3) + 3b(u^2 - v^2) + 3c(u - v) = 0,$$

and therefore

$$a(u^2 + uv + v^2) + 3b(u + v) + 3c = 0,$$

that is,

$$av^2 + (au + 3b)v + au^2 + 3bu + 3c = 0;$$

or, writing herein for v its value,

$$a(\alpha u + \beta)^2 + (\alpha u + \beta)(\gamma u + \delta)(au + 3b) + (\gamma u + \delta)^2(au^2 + 3bu + 3c) = 0;$$

that is,

$$\begin{aligned} & a(\alpha u + \beta)^2 + \alpha\gamma(au^2 + 3bu^2) + \gamma^2(au^4 + 3bu^3 + 3cu^2) \\ & + (\alpha\delta + \beta\gamma)(au^2 + 3bu) + 2\gamma\delta(au^3 + 3bu^2 + 3cu) \\ & + \beta\delta(au + 3b) + \delta^2(au^2 + 3bu + 3c) = 0; \end{aligned}$$

or, reducing by the equation $au^3 + 3bu^2 + 3cu + d = 0$, this is

$$\begin{aligned} & a(\alpha u + \beta)^2 + \alpha\gamma(-3cu - d) + \gamma^2(-du) \\ & + (\alpha\delta + \beta\gamma)(au^2 + 3bu) + \beta\gamma\delta(-d) \\ & + \beta\delta(au + 3b) + \delta^2(au^2 + 3bu + 3c) = 0, \end{aligned}$$

and, collecting the terms, this is

$$\begin{aligned} & u^2 a (\alpha^2 + \alpha\delta + \delta^2 + \beta\gamma) \\ & + u \{a(2\alpha\beta + \beta\delta) + 3b(\alpha\delta + \delta^2 + \beta\gamma) - 3c\alpha\gamma - d\gamma^2\} \\ & + a\beta^2 + 3b\beta\delta + 3c\delta^2 + d(-\alpha\gamma - 2\gamma\delta) = 0. \end{aligned}$$

We can, from this equation, and the equation $au^3 + 3bu^2 + 3cu + d = 0$, eliminate u , thus obtaining a relation between $a, b, c, d, \alpha, \beta, \gamma, \delta$; and, this relation being satisfied, the two equations then determine u rationally in terms of these quantities.

We may without loss of generality assume $\alpha\delta - \beta\gamma = 1$; and, this being so, if we then further assume $\alpha + \delta = -1$, then we have

$$\alpha^2 + \alpha\delta + \delta^2 + \beta\gamma = 0,$$

which is, as above appearing, the condition for $\phi^3 x = 0$. The foregoing equation in u thus becomes

$$\begin{aligned} & u \{a\beta^2(\alpha - 1) - 3b\alpha^2 - 3c\alpha\gamma - d\gamma^2\} \\ & + \{a\beta^2 + 3b\beta\delta + 3c\delta^2 - d\gamma(\delta + 1)\} = 0; \end{aligned}$$

a linear equation giving (in a simplified form) the like results to those given by the quadric equation; *viz.* substituting in the cubic equation the value of u given by the linear equation, we have a relation between $a, b, c, d, \alpha, \beta, \gamma, \delta$; and, this relation being satisfied, u has the determinate value given by the linear equation.

The only way in which u can cease to have this determinate value, and so be capable of being any root whatever of the cubic equation, is when the linear equation becomes $0 = 0$; *viz.* if

$$\begin{aligned} & a\beta^2(\alpha - 1) - 3b\alpha^2 - 3c\alpha\gamma - d\gamma^2 = 0, \\ & a\beta^2 + 3b\beta\delta + 3c\delta^2 - d\gamma(\delta + 1) = 0, \end{aligned}$$

equations which are, in fact, satisfied by the foregoing values of $\alpha, \beta, \gamma, \delta$, as may be verified without difficulty.

It is to be remarked that if, instead of the root u and the equation $v = \frac{\alpha u + \beta}{\gamma u + \delta}$, we consider the root v and the equation $u = \frac{-\delta v - \beta}{-\gamma v + \alpha}$, then, instead of $\alpha, \beta, \gamma, \delta$, we have $\delta, -\beta, -\gamma, \alpha$, and the corresponding equations are

$$\begin{aligned} & a\beta^2(\delta - 1) - 3b\delta^2 + 3c\gamma\delta + d\gamma^2 = 0, \\ & a\beta^2 + 3b\alpha\beta + 3c\alpha^2 + d\gamma(\alpha - 1) = 0, \end{aligned}$$

equations which are also satisfied by the foregoing values of $\alpha, \beta, \gamma, \delta$. And the four equations, together with $\alpha\delta - \beta\gamma = 1$ and $\alpha + \delta = -1$, are more than sufficient to determine the foregoing values of $\alpha, \beta, \gamma, \delta$.

But we further verify without difficulty that the foregoing values of $\alpha, \beta, \gamma, \delta$ give identically

$$(a, b, c, d \mid \alpha x + \beta y, \gamma x + \delta y)^3 = -(a, b, c, d \mid x, y)^3,$$

or the formulæ lead to an automorphic transformation of the binary cubic $(a, b, c, d \mid x, y)^3$. And conversely, starting from the notion of the automorphic transformation of the binary cubic, we ought to be able to obtain the foregoing formulæ.

For greater convenience, I write the equation of transformation in the form

$$(a, b, c, d \mid \alpha x + \beta y, \gamma x + \delta y)^3 = -\theta(a, b, c, d \mid x, y)^3;$$